

Electromagnetic Vibration and Noise of the Permanent-Magnet Synchronous Motors for Electric Vehicles: An Overview

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Abstract—Permanent-magnet synchronous motors (PMSMs) have been widely used as the driving core in electric vehicles due to the high power/torque density and high efficiency. Recent years have witnessed a great effort on the vibration and noise of PMSMs in order to ensure the driving comfort. In this paper, the state of the art and recent progress of the electromagnetic vibration and noise of PMSMs for electric vehicles are presented, with emphasis on the induced mechanism, prediction method, suppression method, and sound quality. The future trends faced are also described.

Index Terms—Electromagnetic vibration and noise, induced mechanism, permanent-magnet synchronous motors (PMSMs), prediction method, sideband noise, sound quality, suppression method.

I. INTRODUCTION

THE rapid growth of car ownership leads to the energy shortage and environmental pollution, and the development of electric vehicles is an effective way to improve the above problems. Permanent-magnet synchronous motors (PMSMs) have been widely used to drive the electric vehicles due to the high power/torque density and high efficiency [1]. As the PMSM replaces the engine, the motor noise has become one of the major noise sources of electric vehicles. In order to ensure the driving comfort, the vibration and noise characteristics of the PMSM deserve our attention [2].

The mechanical, aerodynamic, and electromagnetic origins are the main sources of vibration and noise in PMSMs. Mechanical noise is mainly caused by the mechanical factors such as bearing. The general frequency is relatively low, and its source identification technology is explicit. Aerodynamic noise mainly comes from the cooling fan. However, the PMSMs in electric vehicles are normally free of fan with air cooling or water cooling. As a result, the contribution of aerodynamic noise can be ignored. Electromagnetic noise is mainly induced due to the normal electromagnetic force acting on the surface of stator teeth and permanent magnets (PMs), which is closely related to the electromagnetic parameters and

control strategy of the motors. The electromagnetic noise is the main constitution in most PMSMs. Therefore, the study on the electromagnetic vibration and noise is of great significance for the noise reduction of driving motors.

The purpose of this paper is to overview the state of the art and recent progress of the electromagnetic vibration and noise of PMSMs with different magnetic flux paths. On this basis, the induced mechanism, prediction method, suppression method, and sound quality of the vibration and noise in PMSMs will be emphasized. This paper provides guidance for the design of low-noise PMSMs for electric vehicles.

This paper is structured as follows: In Section II, the major topologies of the PMSMs for electric vehicles are described, and the corresponding induced mechanism of the electromagnetic vibration and noise is discussed. In Section III, the prediction method of vibration and noise for PMSMs will be discussed in detail. The optimization methods from the perspective of excitation source and modal parameters are introduced in Section IV. The suppression methods of the sideband tone noise caused by the switching of the inverter are also discussed in Section V. In Section VI, the state of the art of the sound quality is discussed. Finally, a conclusion is drawn in Section VII.

II. MECHANISM OF THE ELECTROMAGNETIC VIBRATION AND NOISE

Based on the directions of magnetic flux paths, the PMSMs can be classified into radial-flux, axial-flux, linear-flux, and transverse-flux motors [1]. Currently, the majority of the PMSMs used in electric vehicles are radial flux. However, axial-flux motors have drawn a grown concern in the field of electric-drive system and are endowed with a very broad prospects due to the compact structure and high torque/power density. Koenigsegg Regera, for example, is equipped with three axial-flux PMSMs from YASA Motors. In the following, the discussion will be concentrated on the vibration and noise of the above-mentioned two topologies, namely, the radial-flux and axial-flux motors, which can be extended to other topologies. The schematic of the radial-flux and axial-flux PMSMs are depicted in Fig. 1.

The vibration and noise of the electric motors are mainly caused by the normal electromagnetic force acting on the surface of stator teeth and PM. The torque ripple caused by the tangential electromagnetic force can also bring considerable vibration and noise in some cases, which is closely related to

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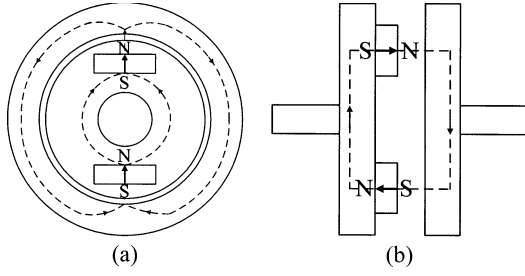


Fig. 1. Schematic of the PMSMs [1]. (a) Radial flux. (b) Axial flux.

the mounting conditions of the PMSMs. As illustrated in [3], remarkable lateral and torsional motion of the motor will be induced when the mounting stiffness is low. Under this circumstance, the structure-borne noise from the electric axes can be present or even dominating over the direct airborne radiated noise of the housing. As this kind of vibration and noise can be eliminated by increasing the mounting stiffness and reducing the torque ripple, this paper focuses on the airborne radiated noise of the housing caused by the normal electromagnetic force.

The normal electromagnetic force can be calculated from the Maxwell tensor method

$$P_n = \frac{B_n^2 - B_t^2}{2\mu_0} \quad (1)$$

where B_n and B_t denote the normal and tangential flux densities, respectively. P_n is the normal electromagnetic force density. μ_0 is the permeability of vacuum.

The air-gap flux density fluctuates periodically with the time and space; thus, clear time-space distribution features can be found in electromagnetic force as well. Generally, the electromagnetic force can be expressed as a superposition of a series of rotating waves with different frequencies and spatial order, that is,

$$P_n(\theta, t) = \sum_m P_m \cos(\omega_m t - m\theta + \varphi_m) \quad (2)$$

where P_m is the amplitude of the electromagnetic force with specific spatial order m , angular frequency ω_m , and phase angle φ_m .

There are two essential conditions under which the resonance occurs, and large vibration and noise are generated in PMSMs. First, the spatial order of the electromagnetic force should be consistent with the circumferential order of the natural mode. Second, the frequency of the force moves close to the corresponding modal frequency. The main noise radiator of radial-flux motors is the annular housing, while that of axial-flux motors is the disk-shaped cover. A gap exists between the modal characteristics of these two kinds of PMSMs, which leads to obvious discrepancy in the mechanism of vibration and noise.

A. Radial-Flux PMSMs

The vibration and noise of radial-flux PMSM are induced by the radial electromagnetic force. Fig. 2 shows the distribution of the electromagnetic force with different spatial orders at a certain time.

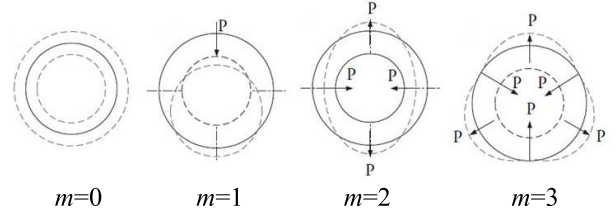


Fig. 2. Distribution of the radial electromagnetic force [4].

According to [5], the deformation of the stator can be expressed as

$$\begin{aligned} Y_0 &= -\frac{r_i N}{Eh} \sigma_{\text{rad},0} \\ Y_1 &= \frac{4}{3} \frac{r_i l}{E \left(\frac{d}{L}\right)^4 L} \sigma_{\text{rad},1} \\ Y_{m \geq 2} &= Y_0 \left(\frac{2\sqrt{3}N}{h} \frac{1}{m^2 - 1} \right)^2 \sigma_{\text{rad},m} \end{aligned} \quad (3)$$

where Y_m is the amplitude of the stator deformation, σ is the radial electromagnetic force density, E is Young's modulus of the stator, r_i and N are the inner and average radii of the stator yoke, respectively. h is the radial thickness of the stator yoke, l is the axial length of the yoke, d is the shaft diameter, and L is the distance of the bearings.

It is shown in (3) that the vibration level is proportional to the amplitude of the electromagnetic force and inversely proportional to the fourth power of the spatial order. Therefore, the electromagnetic force with lower spatial order plays a leading role in the generation of vibration and noise of PMSMs.

The PMSMs can be classified into the integer-slot and the fractional-slot motors. The integer-slot PMSMs are commonly applied in the centralized-driving powertrain. The typical integer-slot traction PMSMs have 4–6 pole pairs, and the spatial order of the lowest nonzeroth radial electromagnetic force is greater than eighth. As the corresponding modal frequencies of these high-order modes are usually above 10 kHz and the vibration is inversely proportional to the fourth power of the spatial order, the contribution of the high-order force to the vibration and noise is weak. However, for the automotive-size PMSM, the frequency of the zeroth mode is around 5 kHz, which is sensitive to the human ear [6]. This leads to the zeroth-order force instead of the lowest nonzeroth force being the major vibration source for this kind of PMSMs [7].

Fractional-slot PMSMs are also widely used, such as the hub motors. For the commonly used fractional-slot PMSM with double-layer winding, the spatial order of the lowest nonzeroth force is equal to the greatest common divisor of the pole and slot numbers. Compared with integer-slot motors, the spatial order of electromagnetic force in fractional-slot motors is lower. The vibration is inversely proportional to the fourth power of the spatial order, which means that the lower spatial order can produce more prominent vibration and noise. Therefore, the lowest nonzeroth-order force is decisive for the vibration and noise in fractional-slot motors [8]–[10]. Besides, as the modal frequency of the zeroth mode is relatively higher than the lowest nonzeroth mode for the

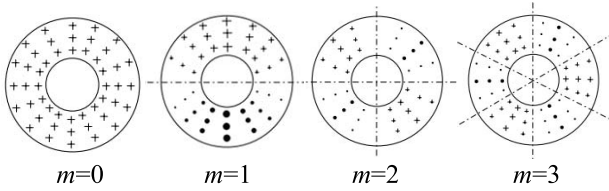


Fig. 3. Distribution of the axial electromagnetic force [13].

automotive-size PMSMs, the contribution of the zeroth force to the vibration and noise is negligible [11], [12].

B. Axial-Flux PMSMs

The vibration and noise of axial-flux PMSMs are induced by the axial electromagnetic force. Fig. 3 shows the distribution of electromagnetic force with different spatial orders at a certain time.

It can be seen from Fig. 3 that along the circumferential direction at a certain radius, the axial electromagnetic force also exhibits the spatial periodicity. As the disk-shaped cover is the main noise radiator of axial-flux PMSM, whose zeroth mode has the lowest natural frequency; it is prone to resonate with the zeroth-order force [14]. This making the vibration and noise generation mechanism of axial-flux PMSM differs from that of radial-flux PMSM, along with some novel features.

- 1) Compared with radial-flux motors, the zeroth-order force is the main source of the vibration and noise in axial-flux motors. As a result, pole and slot combinations with low-noise level in radial-flux motors may induce high-noise level for axial-flux motors [14]–[16].
- 2) A reverse relationship exists between the vibration level and the torque pulsation in radial-flux motors, which means that low-torque pulsation cannot guarantee low vibration and noise [8], [17]. However, as the source of torque pulsation and zeroth force is identical, low-torque pulsation can ensure low vibration and noise for axial-flux motors, which is a huge superiority compared with radial-flux motors [16].

C. Internal- and External-Rotor PMSMs

In addition to the internal-rotor topology, the PMSM can also be designed as the external-rotor structure, as shown in Fig. 4. The internal-rotor PMSMs are commonly used as centralized-driving motors, while the external-rotor PMSMs can be directly connected with hub served as the in-wheel motors.

The vibration and noise of internal-rotor motors come from the electromagnetic force acting on the surface of stator teeth, while that of external-rotor motors come from the electromagnetic force acting on the surface of PMs. Due to the differences in the frequency characteristics of the force, the vibration and noise characteristics of internal- and external-rotor motors are also different [18]. The frequency of the electromagnetic force acting on the stator teeth is $2kf_1$. While for the external-rotor PMSM, the frequency of electromagnetic force is $uQ_s f_r$ under no load and $3kN_t f_r$ or $6kN_t f_r$ under load, which depends on the pole and slot combinations [16], [19].

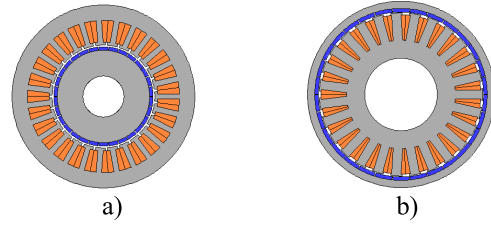


Fig. 4. Schematic of the PMSM. (a) Internal rotor. (b) External rotor.

Consequently, the frequency of the noise radiated by the external-rotor PMSMs differs from that of the internal-rotor PMSMs. Here, f_1 is the fundamental current frequency, f_r is the rotation frequency, Q_s is the number of slots, and N_t is the spatial period of the motor.

Moreover, a special acoustic phenomenon can be observed for the external-rotor PMSMs. Since the acoustic source rotates with the same speed as the motor, the Doppler effect will be produced, which will lead to a certain offset of the temporal order of the electromagnetic noise [20].

III. PREDICTION OF THE ELECTROMAGNETIC VIBRATION AND NOISE

An accurate prediction model of the vibration and noise is desperately needed in order to study the vibration and noise characteristics of the PMSMs. The calculation of the electromagnetic vibration and noise is a complex multiphysics issues, including the electromagnetic field, structure mode, vibration response, and acoustic radiation, which has always been a research focus. As the electromagnetic force, natural mode, vibration, and noise can be calculated by the numerical or analytical method, three main methods are commonly used:

- 1) numerical method;
- 2) analytical method;
- 3) semianalytical method;

A. Numerical Method

The numerical method is to predict the vibration and noise of the PMSMs through the finite-element (FE) software. The prominent advantage of the numerical method lies in that it can take into account the complex shapes and installation conditions of the motors, and thus acquiring a higher precision.

Fig. 5 shows the general flowchart of the vibration and noise calculation. The numerical method can integrate the control model to obtain the phase current of each phase under various operating conditions by the given torque and rotation speed. The output current serves as the input of the electromagnetic model for electromagnetic force calculating, which can be performed with the commercial electromagnetic analysis software, such as JMAG and Maxwell.

Subsequently, the structural model is established to capture the modal characteristics of the motor. As the PMSM is an assembly composed of many parts, considerable errors could be brought if the stator is considered only. Thus, the entire FE model of the PMSM is strongly recommended for the structural analysis. With the entire FE model, the mass and

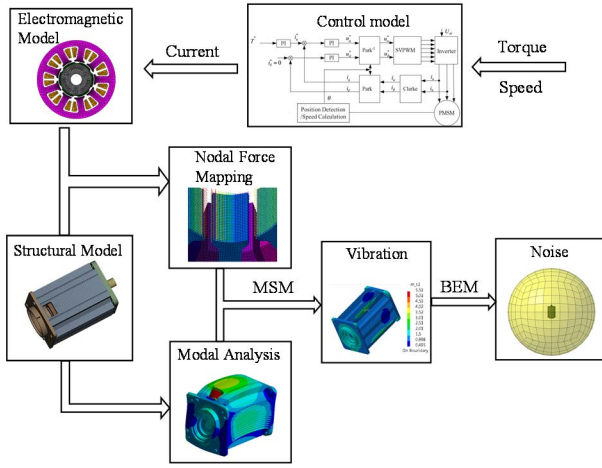


Fig. 5. General flowchart of the vibration and noise calculation with the numerical method.

stiffness effect of each part to the modal characteristics can be taken into account. The contact conditions of different parts can be set up directly with commercial software, like ANSYS. The focus includes the pretension force between the stator and the housing brought by the interference fit and bolts, and the rotational degree of freedom between the inner and outer rings of the bearings [21]. Besides, the real installation condition can be considered using the entire FE model. With the entire FE model, we can add the constraints to the model directly according to how the PMSM is fixed on the platform. On the contrary, the single stator model can only be constrained with both ends simply supported or clamped, which is obviously different from the real installation.

Finally, with the electromagnetic force exerted in the surface of the stator teeth, the vibration response and acoustic radiation of the PMSM can be calculated through the modal superposition method and boundary element method, respectively. LMS Virtual. Lab can be used for the vibroacoustic analysis.

In order to estimate the accuracy of the predicted results, it is necessary to distinguish the mechanical and aerodynamic noises from the measured results. The aerodynamic noise can be eliminated by removing the fans during the noise test. For the air-cooled and water-cooled PMSM, the contribution of the aerodynamic noise can be ignored. Subsequently, the mechanical noise can be recognized based on the sudden power-OFF method. With the power OFF, the electromagnetic noise is eliminated and the mechanical noise can be distinguished directly. The accuracy of the predicted results can be estimated by filtering out the mechanical and aerodynamic noises from the measured results.

There are two major difficulties in the prediction process of the vibration and noise in Fig. 5, which are the main factors that affect the accuracy of the simulated noise. One is the acquisition of the equivalent orthotropic material parameters of the stator core and winding. The other is the loading of the distributed electromagnetic force. The solution of these two problems is the key to accurately predict the electromagnetic noise

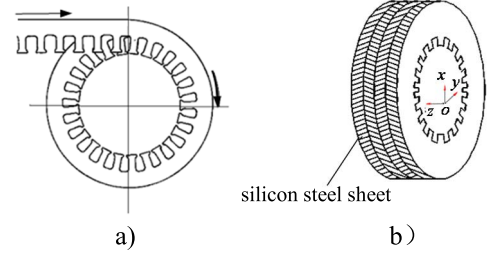


Fig. 6. Schematic of the laminated stator core. (a) Processing. (b) Stacking method of the silicon steel sheet.

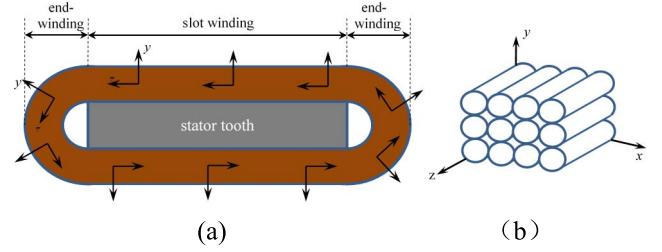


Fig. 7. Schematic of the winding [22]. (a) Cross section. (b) Formation of coils.

1) *Acquisition of the Equivalent Orthotropic Material Parameters of the Stator Core and Winding:* The stator core of the radial-flux PMSM is fabricated by stacking the silicon steel sheet along the axial direction, and the winding is constituted by the coils wound on the stator teeth, as shown in Figs. 6 and 7, respectively.

Therefore, the mechanical properties of the actual stator core and winding in different directions are not the same. If we establish the stator core and winding with many silicon steel sheets and coils in FE software, the computation will face huge challenges, and the contact is also difficult to accurately simulate. The general idea is to establish the solid model of the stator core and winding. The stacking effect can be considered by adopting the equivalent orthotropic material parameters, which contains Young's modulus, shear modulus, and Poisson's ratio. If the isotropic material parameters are used, certain errors will be brought to the accuracy of the model [23], [24]. While adopting the equivalent orthotropic material parameters can improve the modeling accuracy greatly [21], [25].

In the following, the acquisition method of the equivalent orthotropic materials parameters for radial-flux motors will be discussed in detail. This method can also be extended to axial-flux motor and other kinds of motors.

There are two main methods to identify the equivalent orthotropic materials parameters. One is to measure the material parameters directly by the ultrasonic method [26], but this method requires specific experimental equipment. The other one is to approximate the tested modal frequencies by adjusting the orthotropic material parameters in the FE model.

Since Poisson's ratio has a relatively small influence on the modal frequencies, it is generally considered to be consistent in all the three directions in order to simplify the analysis. The equivalent orthotropic material parameters to be acquired include Young's modulus in the x -, y -, and z -directions (E_x , E_y , and E_z) and the shear modulus in the xy , yz , and

TABLE I
ORTHOTROPIC PARAMETERS OF THE STATOR CORE AND THE WINDING

Material parameter	silicon steel sheet	Stator core	Copper	Winding
$\rho(\text{kg/m}^3)$	7850	7480	8300	5086
ν	0.3	0.3	0.3	0.3
E/GPa	$E_w = 210$	$E_x = E_y = 178.5 = 85\%E_w$ $E_z = 210 = 100\%E_w$	$E_c = 120$	$E_x = E_y = 0.6 = 0.5\%E_c$ $E_z = 120 = 100\%E_c$
G/GPa	$G_w = 80.77$	$G_{xy} = 68.65 = 85\%G_w$ $G_{yz} = G_{xz} = 11 = 13.6\%G_w$	$G_c = 46.15$	$G_{xy} = 0.4615 = 1\%G_c$ $G_{yz} = G_{xz} = 0.923 = 2\%G_c$

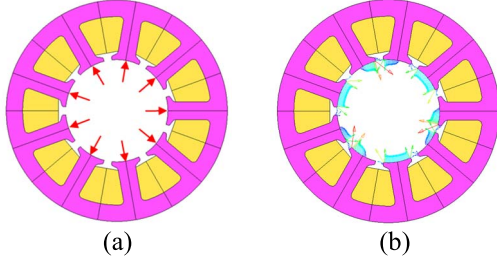


Fig. 8. Loading method of the electromagnetic force. (a) Concentrated force. (b) Distributed force.

xz plane (G_{xz} , G_{yz} , and G_{xy}). As shown in Figs. 6 and 7, the silicon steel sheet is stacked along the axial direction (z -axis), and the coils are formatted along the axial direction as well. Due to the symmetry of the structure, the material properties are assumed the same in the x - and y -directions, namely, $E_x = E_y$ and $G_{xz} = G_{yz}$. This means merely that the four parameters of Young's modulus and the shear modulus are independent [22]. E_x and G_{xy} have great influence on the axial modes $m = 0$, and G_{xz} has great influence on the axial modes $m = 1$, while E_z has a little influence on the modal frequencies. Table I lists the orthotropic material parameters of the stator core and winding for a PMSM, which can provide reference for the modal modeling of the electric motors.

2) *Loading the Distributed Electromagnetic Force*: As the main source of motor vibration and noise, the electromagnetic force has an important influence on the accuracy of the numerical prediction method. The electromagnetic force is equivalent to a concentrated force exerted in the center of stator teeth or PM in [8], [10], and [27]–[30]. However, the actual force is a distributed force, as shown in Fig. 8.

On the one hand, the concentrated equivalent method ignores the electromagnetic force with higher spatial order. According to the sampling theorem, the maximum spatial order of the electromagnetic force that can be considered by the concentrated equivalent method is

$$n = \frac{Z}{2} \quad (4)$$

where Z is the number of stator teeth for the internal-rotor motors or PMs for the external-rotor motors. According to (4), the concentrated equivalent method can only take into account the electromagnetic force below fourth for the PMSM with nine slots in Fig. 8. As a result, this method will bring considerable error, especially for the motor with few pole and slot numbers. On the other hand, the concentrated equivalent method neglects the uneven distribution of the force, resulting

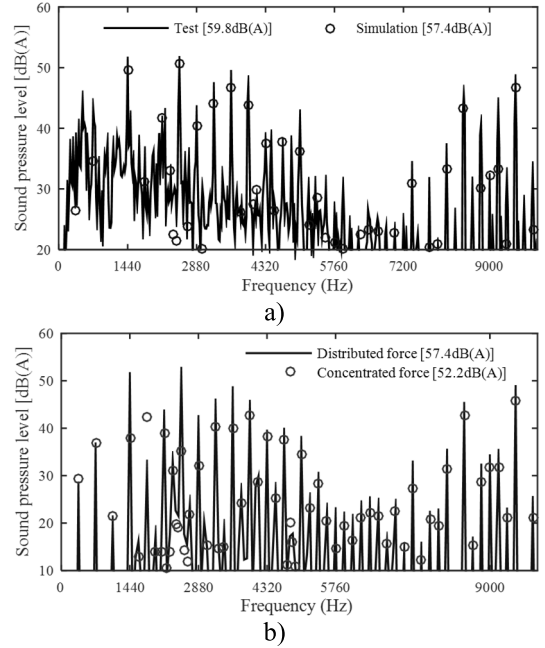


Fig. 9. Electromagnetic noise of a PMSM with six poles and nine slots under rated load [31]. (a) Comparison of the tested and simulated noise spectra with distributed force. (b) Comparison of the simulated noise spectrum with the concentrated force and distributed force.

a large error in the amplitude of the loaded force with different spatial orders.

In [14], [16], [19], and [31], a nodal force transfer method is proposed to consider the uneven distribution of the electromagnetic force. The nodal force calculated by the electromagnetic model with JMAG is mapped to the structural model following the interpolation law through the commercial software LMS Virtual. Lab; thus, the uneven distribution of the electromagnetic force can be taken into account. As shown in Fig. 9, this method gives better accuracy than the concentrated equivalent method.

The numerical method is more accurate than the analytical or the semianalytical method, but it consumes more time. Especially for axial-flux PMSMs, the 3-D FE model is essential for force calculating, resulting in more time consumed to obtain the vibration and noise. Another disadvantage of the numerical method is that it is difficult to predict the vibration and noise of the PMSMs over a wide speed range, which is also an urgent problem to be solved.

B. Analytical Method

As the numerical method consumes a considerable amount of time and fails to simulate the vibration and noise over a

wide speed range, many scholars predict the vibration and noise from the perspective of analytical method. The radiated sound power of a motor can be expressed as

$$\Pi = \sum_{m,n=0}^{\infty} \sigma_{mn} \rho_0 c_0 S \langle \overline{v_{mn}^2} \rangle \quad (5)$$

where σ_{mn} is the sound radiation efficiency, ρ_0 is the air density, c_0 is the sound speed, S denotes the area of the noise radiating surface, and $\langle \overline{v_{mn}^2} \rangle$ is the spatial averaged mean-square velocity.

From (5), we can see that analytical modeling of the vibration and noise of a PMSM requires to accurately calculate the surface vibration velocity and the sound radiation efficiency, while the surface vibration velocity is related to the electromagnetic force and the inherent modal characteristics. The analytical model of the electromagnetic field is not the focus of this paper and will not be elaborated here. The analytical methods of the radial-flux and axial-flux PMSMs are not identical due to their different noise radiators.

1) *Radial-Flux PMSM*: The main noise radiator of radial-flux motors is the annular shell. In order to predict the vibration and noise, the modal shapes and frequencies should be first calculated and then the sound radiation efficiency.

In the field of modal analysis, the stator is equivalent to a ring in [32] and [33]. However, the ring model is only suitable for the stator with smaller aspect ratio, and the influence of winding, housing, and frame is neglected. Afterwards, the finite-length cylindrical shell model gradually replaces the ring model to calculate the modal parameters of the radial-flux PMSMs, as shown in Fig. 10. The finite-length cylindrical shell can take into account the influence of the end effects, which is especially suitable for the motors with closed cylindrical structure [4]. Recently, the finite-length cylindrical shell model is improved by considering the orthotropic material parameters of the stator. The proposed method can achieve better precision. Unfortunately, this method ignores the influence of the housing and the frame; thus, there is room for further improvement [34].

Sound radiation efficiency of the radial-flux motor can be calculated by the spherical model, the infinite-length model and the finite-length model [35], [36]. The spherical model is only appropriate for the PMSMs with an aspect ratio approaching unity, which could be in severe error for the motors with large aspect ratio. The infinite-length model cannot reflect the influence of the aspect ratio and also ignores the end effect. Actually, most of the PMSMs are not infinitely long in the axis direction and the end effect is significant. Therefore, the finite-length model is widely used to calculate the vibration and noise currently.

2) *Axial-Flux PMSM*: Different from the radial-flux motors, the stator or the outer rotor of axial-flux motor is directly connected with the cover, resulting in the main noise radiator is the disk-shaped cover rather than the cylinder-shaped housing. Thus, the analytical model for radial-flux motors cannot be applied to axial-flux motors. At present, there are few literatures focusing on the analytical prediction of the vibration and noise for axial-flux motors. In [13], the electromagnetic

vibration and noise of an axial-flux PMSM are predicted analytically. As the analytical model is a lumped parameter model and the sound radiation efficiency is considered to be unity in the entire frequency band, its accuracy is not satisfied. A circular plate model is proposed in [15] to calculate the vibration and noise of the axial-flux motors. This method cannot only predict the surface velocity at an arbitrary point of the cover with an improved precision but also can calculate the vibration and noise over a wide speed range, which provides the guidance for the analytical modeling of the vibration and noise for axial-flux motors.

In general, the analytical methods can calculate the vibration and noise fastly over a wide speed range, but there are also some shortcomings.

- 1) The actual boundary conditions of electric motors are complicated. However, the analytical method can only describe the supporting state of the motor by three kinds of boundary conditions, namely, free, simple, and fixed. This leads to a relatively low precision of the analytical model.
- 2) The analytical method is only appropriate to motors with regular configuration. While for the motors with complex structure, a severe error could be induced.

In the future work, a more accurate analytical model can be established by taking into account the housing and end effects together with the orthotropic material parameters.

C. Semianalytical Method

Numerical method has higher accuracy but is time consuming. The precision of the pure analytical method is limited due to the simplification. Therefore, the semianalytical method comes into being. The semianalytical method combines the analytical and numerical methods to predict the vibration and noise of the PMSMs.

As all of the electromagnetic force, structural modes and vibroacoustic can be calculated analytically or numerically, the semianalytical methods are various. A commonly used method is to calculate the electromagnetic force through the FE method or field reconstruction method, the vibration and acoustic radiation can be, then, obtained by the analytical model [37], [38]. However, due to the low precision of the structure modes calculated analytically, this method has the same defects as the analytical method. On this basis, we can replace the analytical modal analysis by a numerical solution to improve the accuracy of the semianalytical model [39]–[41]. Another parameter that affects the accuracy is the sound radiation efficiency. The same idea can be adopted to further improve the precision by calculating the sound radiation efficiency with a numerical solution in the future.

IV. REDUCTION OF THE ELECTROMAGNETIC VIBRATION AND NOISE

The electromagnetic force acting on the surface of the stator teeth or PMs causes the motor to vibrate and radiates noise outward. Therefore, in order to reduce the vibration and noise of the PMSMs, on the one hand, the resonance can be avoided by changing the natural frequency, namely, through the modal programing. On the other hand, it can start from the

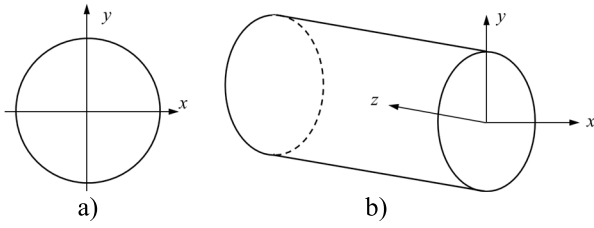


Fig. 10. Equivalent model of the radial-flux model to calculate the modal parameters. (a) Ring model. (b) Finite-length cylindrical shell model.

perspective of reducing the amplitude of the electromagnetic force.

According to [8]–[10], [12], [25], and [42], the motor noise is closely related to the pole and slot combinations. For the conventional radial-flux PMSM, the pole and slot numbers should be selected with higher greatest common divisor to avoid the generation of electromagnetic force with low spatial order. While for the axial-flux PMSM, as the zeroth-order force is important for the vibration and noise, the rule of pole and slot number selections is different from that of radial-flux motors [16]. However, the pole and slot numbers are generally selected in the design stage due to the constraints of motor size and output performance requirement. The vibration and noise can only be reduced by the modal programming or optimizing the amplitude of the electromagnetic force.

A. Modal Programming

Modal programming is to make the modal frequency away from the frequency of electromagnetic force by optimizing the structural parameters. The existing studies focus on the optimization design of the stator structural parameters to increase the stator stiffness, making the modal frequency away from the frequency of the electromagnetic force and thus weakening the vibration and noise [43], [44]. Due to the order features, the frequency of the electromagnetic force varies at different rotational speeds. Therefore, this method is suitable for the fixed-speed motors. However, the PMSMs for electric vehicles are with wide speed range, resulting in the electromagnetic force a broadband excitation. It is difficult to avoid the resonance in all working conditions, thus reducing the vibration and noise from the perspective of modal programming is not very applicable.

B. Reduction of the Amplitude of the Electromagnetic Force

The other reduction method is to optimize the amplitude of the electromagnetic force. First, it is necessary to determine the spatial order and the frequency of the electromagnetic force that contributes most to the vibration and noise, and then optimizing the amplitude of the selected electromagnetic force. There are three main ways that are commonly used.

1) *Optimizing the Electromagnetic Parameters*: Optimizing the electromagnetic parameters is to optimize the structural parameters that are sensitive to the amplitude of the electromagnetic force. The sensitive parameters include the slot opening width, the air-gap length, the pole arc coefficient, and the thickness of the PMs. These structural parameters have little

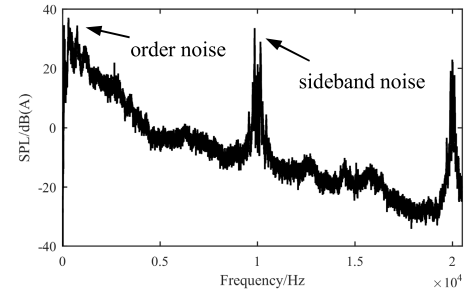


Fig. 11. Tested noise spectrum of a PMSM driving by SVPWM.

influence on the modal frequencies generally [19], [45], [46]. Owing to the genetic algorithms and other multiobjective optimization methods, these parameters can be optimized to reduce the vibration and noise without sacrificing the average output torque. In other words, a well compromised can be achieved between the electromagnetic noise and the torque.

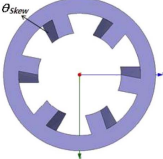
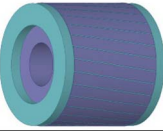
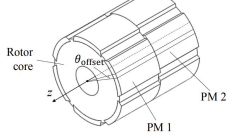
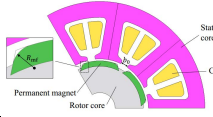
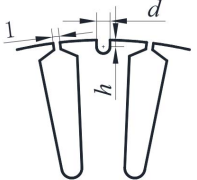
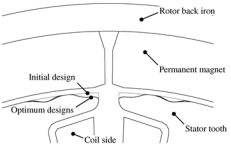
2) *Injecting the Compensation Current*: Injecting appropriate compensation current is another method to suppress the main noise peaks of the PMSM. The armature reaction field induced by the injected current interacts with the PM field, generating the electromagnetic force with the same spatial order and frequency but the opposite phase angle as the original one, so as to reduce the amplitude of the electromagnetic force [12], [47]. The spatial order and frequency of the electromagnetic force that contributes mostly to the radiated noise should be ascertained first with the 2-D fast Fourier transform. On this basis, the amplitude and phase angle of the compensation current harmonic can be determined. This method can weaken the amplitude of the electromagnetic force, thereby reducing the vibration and noise. However, the output torque will be reduced due to the changes of the current angle.

3) *Modification of the Structure*: The vibration and noise of the PMSMs can also be reduced by the modification of the motor structure. One of the widely used approaches is the skewing method, including the stator skewing [48] and rotor skewing [49]. In order to eliminate the axial unbalance force induced by the single-step method, the step-skewed and herringbone-skewed methods are proposed further [50]. Besides, other favored means for noise reduction include the chamfering [50], notching [51], and modification of the tooth shapes [18]. The schematic is shown in Table II. These methods can also be applied to reduce the torque pulsation. These methods are easy to achieve, but inevitably resulting in the decline of the output torque. A balance of the electromagnetic noise and output torque should be taken into account.

V. SUPPRESSION OF THE HIGH-FREQUENCY TONE NOISE

Actually, in addition to the low-frequency order electromagnetic noise mentioned in Section IV, there is another high-frequency sideband tone noise in the PMSMs which is generally driven by the space vector pulsewidth modulation (SVPWM). This kind of noise comes from the sideband electromagnetic force induced by the sideband current due to the inverter switching. Fig. 11 shows the tested noise spectrum of

TABLE II
METHODS AND SCHEMATIC FOR VIBRATION
AND NOISE REDUCTION OF THE MOTORS

Methods	Schematic diagram	Reference
Stator skewing		[48]
Rotor skewing		[49]
Step-skewed		[50]
Chamfering		[50]
Notching		[51]
Tooth shape modification		[18]

a PMSM driving by SVPWM. The carrier frequency is 10 kHz. A considerable sideband noise can be observed at the integral multiples of the carrier frequency.

This kind of high-frequency tone noise is ear piercing. For the PMSMs used in electric vehicles, the electromagnetic noise caused by the modulation strategy is equally deserving our attention in order to ensure the driving comfort. The sideband noise can be reduced by the choice of appropriate switching frequency to avoid the resonance [52]. However, the sideband noise can only be weakened to a certain extent by this method and cannot be eliminated fundamentally.

As the sideband noise is consistent with sideband voltage, the commonly thinking is to spread the voltage spectrum by novel modulation strategies. According to Parseval's theorem, the total energy in the frequency domain is equal to that in the time domain. The sideband voltage harmonics are mainly concentrated around the carrier frequency. If the carrier frequency is changed over time, the amplitude of the voltage harmonics around the original fixed carrier frequency will be reduced and the sideband noise will be weakened as well. These kinds of modulation strategies are the so-called

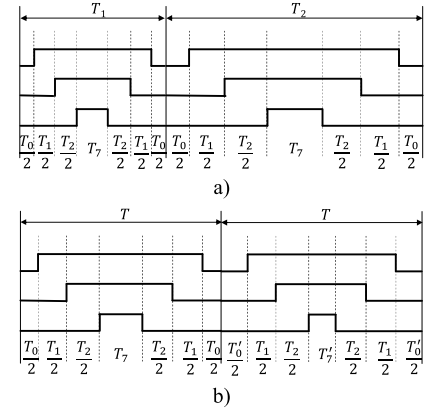


Fig. 12. Time allocation of the voltage vector. (a) RCFM. (b) RPWM.

spread spectrum modulation techniques. The spread spectrum modulation techniques mainly include the random modulation technique and the periodic modulation technique, which are widely used to reduce the electromagnetic interference at the beginning [53] and have been extended to suppress the sideband noise recent years.

A. Random Modulation Technique

The commonly used random modulation techniques contain the random carrier-frequency modulation (RCFM) and the random pulsewidth modulation (RPWM). RCFM is to modulate the carrier frequency by a pseudorandom code, thereby spreading the voltage spectrum [54]. The power spectral density (PSD) of RCFM can be expressed as [55]

$$P(f) = \frac{1}{T_{av}} \left(\frac{V_{dc}}{\pi f} \right)^2 \left\{ E[\sin^2(\pi f DT)] + 2\text{Re} \left\{ \frac{E^2[\sin(\pi f DT)e^{j\pi f t}]}{1 - E(e^{j\pi f t})} \right\} \right\} \quad (6)$$

where T_{av} is the average switching period, V_{dc} is the supply voltage of the inverter, D is the duty ratio, T is the switching period, f is the frequency, and $E(\dots)$ is the statistical expectation.

Generally, the duration of zero vectors T_0 and T_7 in the standard SVPWM are equal. RPWM is to allocate the duration of the zero vectors randomly, thus randomizing the turn-ON instants of the pulse to spread the voltage spectrum [56], [57]. This technique is effective at low modulation indexes. While at high modulation indexes, the spreading effect is not satisfactory. The time allocation of the voltage vectors for the RCFM and RPWM is shown in Fig. 12.

In addition, there are also some improved random modulation techniques. The hybrid random modulation technique can randomize the carrier frequency and the pulsewidth simultaneously [58]. It can spread the noise spectrum with a better effect, but the control strategy is more complex. A novel modulation technique with fixed carrier frequency is also proposed by switching the individual legs more times in a sampling period based on the two random numbers [59]. However, the increase in switching times will be accompanied with a higher inverter loss.

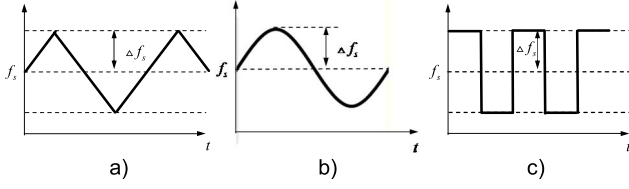


Fig. 13. PCFM. (a) Triangular. (b) Sinusoidal. (c) Square.

One major disadvantage of the random modulation techniques is that it is hard to predict the total harmonic distortion (THD) of current and the switching loss [60]. In addition, with the spreading of the voltage spectrum, the induced-electromagnetic force is broadband. If the selection of the random parameters is not appropriate, resonance may occur and the noise increases on the contrary.

B. Periodic Modulation Technique

The periodic modulation technique refers to the periodic carrier-frequency modulation (PCFM), namely, the carrier frequency changes periodically with time. PCFM includes the triangular carrier-frequency modulation [60], [61], sinusoidal carrier-frequency modulation [62], and the square carrier-frequency modulation [63], as shown in Fig. 13.

The variation law of these three kinds of PCFM with time can be expressed as

$$f_{tri}(t) = \begin{cases} 4\Delta f_s f_m t - f_s & 0 \leq t < \frac{1}{2f_m} \\ -4\Delta f_s f_m t + 3\Delta f_s & \frac{1}{2f_m} \leq t < \frac{1}{f_m} \end{cases} \quad (7)$$

$$f_{sin}(t) = \Delta f_s \sin(2\pi f_m t) \quad (8)$$

$$f_{squ}(t) = \begin{cases} \Delta f_s & 0 \leq t < \frac{1}{2f_m} \\ \Delta - f_s & \frac{1}{2f_m} \leq t < \frac{1}{f_m} \end{cases} \quad (9)$$

where f_m is the period of the frequency variation and Δf_s is the peak instantaneous frequency deviation.

As the carrier frequency of the square carrier-frequency modulation only changes twice in a sampling period, the corresponding spreading effect is not satisfied; thus, the triangular and sinusoidal carrier-frequency modulations are applied more widely. According to [64], the power density of the PCFM is expressed as

$$S_G(f, \beta) = 2 \left| \sum_{n=1}^{\infty} C_n \left\{ \begin{aligned} &J_0(n\beta)\delta(f - nf_s) \\ &+ \sum_{k=1}^{\infty} J_k(n\beta)[\delta(f - nf_s - kf_m) \\ &\quad + (-1)^k \delta(f - nf_s + kf_m)] \end{aligned} \right\} \right|^2 \quad (10)$$

where f_s is the carrier frequency, β is the ratio of Δf_s to f_m , and n and k are the integers.

It can be inferred from (10) that extra voltage harmonics will be induced, whose frequency differs from the original one by kf_m . As a result, the amplitude of the original harmonics is spread, and the tone noise is suppressed. Since the carrier frequency of PCFM at arbitrary time is determined, the THD of current and the switching loss can be calculated. However, this method tends to increase the current THD.

C. Selective Noise Cancellation at Specific Frequency

The random modulation technique and the periodic modulation technique can spread the current spectrum. However, the induced sideband electromagnetic force will be broadband; thus, resonance is prone to occur. The novel strategies that can create a gap in the noise spectrum at specific frequency have been explored to shape the electromagnetic noise.

The thinking of canceling the noise at specific frequency is to build the relationship between the switching period and the duty ratio. It can create a gap in the voltage and noise spectrum. According to (6), the PSD at a specific frequency f_0 and its multiples will be 0 if [55]

$$\sin(\pi f_0 DT) = 0. \quad (11)$$

Thus,

$$T = \frac{k}{f_0 D} \quad (12)$$

where k is an integer.

However, this method is only applicable for removing high-frequency noise that exceeding the human hearing range. A novel modulation strategy is proposed for two universal space vector modulation techniques [65], [66]. The switching period is not only related to the duty ratio but also determined by the switching period in the previous cycle. When the pulses are located at the center of the switching period, it follows [66]:

$$T_{n+1} = \frac{2k - f_0 T_n (1 + D_n)}{f_0 (1 + D_{n+1})}. \quad (13)$$

Compared with the method of (12), this method can eliminate the noise at specific frequency effectively within the human hearing range. This method can also be applied to the conventional SVPWM. Fig. 14 shows the voltage spectrum of a PMSM driven by combining the proposed method of (13) with the standard SVPWM. An obvious gap can be observed at 7 kHz and its multiples as well. Therefore, the method can avoid the resonance effectively within the human hearing range.

VI. SOUND QUALITY

The generally evaluation indicator of the noise for electrical motors is the A-weighted sound power level (SPL). The majority of current studies focus on the evaluation and reduction the A-weighted SPL of the electromagnetic noise. However, there are some drawbacks of the A-weighted SPL.

- 1) The A-weighted SPL is obtained by the superposition of the sound power of the entire frequency band. Two different noise spectra may have the same A-W SPL, but human perception can be totally different. Therefore, the A-W SPL cannot reflect the physiological and psychological of human beings.
- 2) When evaluating the low-frequency and tone noises, a severe error can be induced between the A-W SPL and human perception. However, for the PMSMs driving by the SVPWM, the switching of the inverter will bring about high-frequency tone noise, whose influence on human perception cannot be totally reflected by A-W SPL.

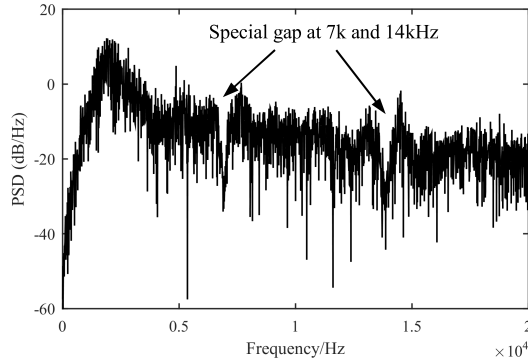


Fig. 14. Voltage spectrum of a PMSM adopted the proposed method.

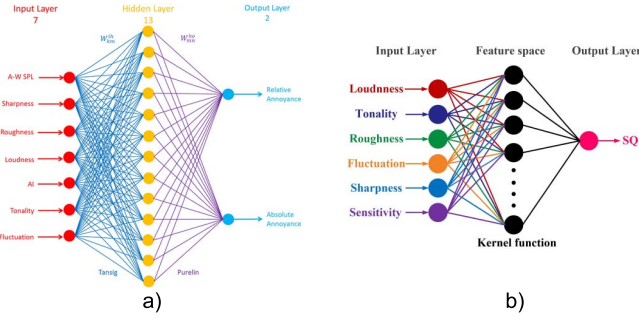


Fig. 15. Sound quality evaluation model. (a) BP neural network model [71]. (b) Support vector regression model [74].

The sound quality, which can reflect the loudness, sharpness, roughness, and other features of the acoustic noise, is more suitable to evaluate the human perception. The motors have replaced the engines to drive the electrical motors, resulting in the motor noise being one of the main noise sources. The sound quality of the PMSMs in electric motors also deserves our attention to ensure the driving comfort.

The research of sound quality contains both the objective evaluation and subjective evaluation. The objective evaluation parameters include the sharpness, roughness, loudness, articulation index, tonality, and the fluctuation. The objective evaluation parameters can reflect the sound quality in some degree, but cannot make a full reflection of human perception. The purpose of objective evaluation is to lay the foundation for the sound quality evaluation model. The computing method of the objective evaluation parameters can be found in [67]. The subjective evaluation is to grade the noise samples by the evaluation subjects with relevant experience. Based on the obtained objective and subjective evaluation results, the sound quality evaluation model of the PMSMs is, then, obtained by establishing the mapping relationship between the subjective evaluation scores and the objective evaluation results. As there is a strong nonlinearity between the subjective scores and objective evaluation parameters, the commonly used sound quality model is the back propagation (BP) neural network or the support vector regression model, as shown in Fig. 15.

At present, there are relatively few studies focusing on the sound quality of electric motors. In [68], a numerical method is proposed to predict the electromagnetic noise of a variable-speed PMSM, and the objective evaluation parameter is calculated. The influence of various modulation strategies

on sound quality is evaluated in [69]. However, none of them establish the universal sound quality evaluation model. In [70]–[72], the BP neural network is selected to obtain the sound quality evaluation model of the PMSMs for electric vehicles. The proposed model is used to diagnose the abnormal noise. Also, the sound quality model is established by the support vector regression model in [73] and [74]. It is pointed out that the sideband noise caused by the switching of the inverter will decorate the sound quality.

The limits of current study and possible improvements on sound quality are as follows.

- 1) Current studies on the sound quality focus on the acoustic noise of motors with uniform velocity and acceleration, which neglect the complex automotive transient operating conditions. In the future work, the sound quality of PMSMs under actual automotive operating conditions should be emphasized to improve the sound quality of electric vehicles.
- 2) The existing researches pay little attention to the optimization design of sound quality of PMSMs. The sideband noise caused by the switching of the inverter deteriorates the sound quality severely. Therefore, we can improve the sound quality of PMSMs by novel modulation strategies, such as the techniques mentioned in Section V.

VII. CONCLUSION

The state of the art and recent progress of the electromagnetic vibration and noise of PMSMs for electric vehicles are overviewed in this paper. The main topologies of PMSMs for electric vehicles are identified first, namely, the radial-flux and axial-flux PMSMs. Then, the pros and cons of current technology are discussed in detail, with emphasis on the induced mechanism, prediction method, suppression method, and sound quality. The discussion aims to reveal the vibroacoustic characteristics of the PMSMs for electric vehicles. In particular, this overview provides a comprehensive guidance for the design of low-noise PMSMs. The conclusions and future trends are as follows.

- 1) The modal force transfer method can give better accuracy than the concentrated equivalent method, which is recommended in the numerical prediction of the vibration and noise for PMSMs. In general, the numerical method is more accurate than the analytical or the semianalytical method, but it consumes more time. Another disadvantage of the numerical method is that it is difficult to predict the vibration and noise of the PMSMs over a wide speed range. A potential solution is the analytical or the semianalytical method.
- 2) The analytical method can be used to predict the vibration and noise over a wide speed range. In the future work, a more accurate analytical model can be established by taking into account the housing and end effects together with the orthotropic material parameters of the stator and winding.
- 3) The spread spectrum modulation technique will increase the THD of the current. As a result, the amplitude of the low-frequency current harmonics, together with

the low-frequency order electromagnetic noise, will also increase. In the future, the influence of the spread spectrum modulation technique on the low-frequency order noise and high-frequency sideband noise should be balanced.

- 4) The prediction of the vibration and noise and the modeling of the sound quality should combine with the complex automotive transient operating conditions. On this basis, the optimization design of sound quality should be emphasized.

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